

# GenMol UDLM engineering exploration

engineering-v9-ct-ce-objectives | complete | 2026-09-07T11:22:42.344741+00:00

## Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

CT and clean CE share the training settings recorded in this protocol. Equal updates and examples do not imply equal compute or independently optimized objectives; training losses have different scales and targets.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED  $\geq$  0.6 and SA  $\leq$  4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

## Prospective sampling design

`{"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "min_add_len": 40, "objective_comparison": {"contrast": "CE minus CT for each seed, then equal-seed mean; report both contrasts even when negative.", "primary": {"control_config": "ct_t100", "temperature": 1.0, "treatment_config": "ce_t100"}, "secondary": {"control_config": "ct_t050", "temperature": 0.5, "treatment_config": "ce_t050"}, "selection_history": "V5/V6 outcomes and V7 throughput were known before design; temperature 0.5 is an informed secondary engineering setting."}, "parameterization_controls": "Convert CE clean-denoiser logits to LOO at the current noisy state/time before temperature and top-p; CT logits already parameterize raw LOO.", "predictor_transitions_per_molecule": 128, "randomness": 0.0, "raw_loo_top_p": 1.0}`

The original V8 CT controller and campaign remain failed. This comparison uses its separately audited checkpoint plus a new completed V8b CE run; never label V8 campaign success.

Training source revisions differ through controller/reporting changes; model/training implementation and resolved common training settings are matched by recorded hashes.

128000 training exposures are configured accounting, not distinct molecules or an independent live row trace. Finite final tensors do not establish finiteness of every intermediate update.

V5/V6 informed this engineering design, especially temperature 0.5; this is not independent confirmation or a superiority claim.

Two seeds and 100 requests per setting are small; pairing is by seed/run, not molecule-level trajectories or independence of generated observations.

Report validity, uniqueness, quality and diversity under both strict SAFE decoding and released repair/largest-component selection. Quality counts unique QED $\geq$ 0.6 and SA $\leq$ 4 molecules divided by requests.

Equal updates/examples and shared hyperparameters do not imply equal compute or independently optimized CT/CE objectives. The baseline uses 50k MDLM updates and three 1000-request seeds.

Final UDLM seeds 0/1/2 remain reserved. No automatic generation follows this materialization, and no model promotion or longer training follows the results.

## Objective comparison

CT / raw-LOO predicts clean leave-one-out probabilities. Clean CE predicts clean denoiser probabilities, converted to LOO logits at the current noisy state and time before temperature or top-p. The conversion adds no backbone evaluation. Shared sampling settings do not imply independently optimized objectives. Objective labels and checkpoint evidence below describe completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate\_grad\_batches": 4, "arms": {"CE": {"checkpoint\_sha256": "b7d674ed5bddd1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "controller\_status": "completed", "launch\_receipt": {"relative\_path": "output/udlm/engineering\_v8b/ce\_e\_1000\_b128\_w2/launch\_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size\_bytes": 17020}, "protocol": {"relative\_path": "experiments/udlm/protocols/engineering\_v8b\_ce\_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b"}, "request\_receipt": {"relative\_path": "output/udlm/engineering\_v8b/ce\_e\_1000\_b128\_w2/request\_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size\_bytes": 8621}, "resolved\_config\_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source\_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal\_receipt": {"relative\_path": "output/udlm/engineering\_v8b/ce\_e\_1000\_b128\_w2/terminal\_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size\_bytes": 11719}, "udlm\_denoiser\_metadata": {"inference\_conversion": "subtract\_local\_forward\_log\_likelihood\_before\_controls", "objective": "clean\_token\_cross\_entropy", "parameterization": "x0\_denoiser", "schema\_version": 1}}, "CT": {"checkpoint\_sha256": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "checkpoint\_status": "separately\_validated\_after\_controller\_failure", "controller\_status": "failed", "post\_exit\_audit": {"relative\_path": "output/udlm/engineering\_v8/ct\_ce\_e\_1000\_b128\_w2/post\_exit\_audit/post\_exit\_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size\_bytes": 9166}, "resolved\_config\_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source\_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal\_receipt": {"relative\_path": "output/udlm/engineering\_v8/ct\_ce\_e\_1000\_b128\_w2/ct/terminal\_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size\_bytes": 9822}}, "common\_mask\_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning\_variant": "film\_adaln", "example\_exposures\_per\_arm": 128000, "global\_batch\_size": 128, "gpu\_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched\_common\_config\_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro\_batch\_size": 16, "optimizer\_updates": 1000, "original\_failed\_campaign": {"relative\_path": "output/udlm/engineering\_v8/ct\_ce\_e\_1000\_b128\_w2/campaign\_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size\_bytes": 846}, "original\_v8\_protocol": {"relative\_path": "experiments/udlm/protocols/engineering\_v8\_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak\_learning\_rate": 0.0003, "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "scheduler": {"decay\_floor\_lr": 3e-06, "horizon\_updates": 1000, "name": "half\_cosine\_with\_linear\_warmup\_and\_floor", "warmup\_updates": 50}, "training\_implementation\_hashes\_sha256": "c9641b7e447d1f4617599e4df338fde0010fe68f7877566de37d2e6c5a6ccd2b", "training\_seed": 1500, "uniform\_mixture": 0.0002}

## Reference means (context only)

| Reference              | validity | uniqueness | quality  | diversity |
|------------------------|----------|------------|----------|-----------|
| local_mdml_50000       | 1.000000 | 0.998667   | 0.858000 | 0.823021  |
| paper_genmol_v1_table1 | 1.000000 | 0.997000   | 0.846000 | 0.818000  |

## Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

| Config  | Seed | Total NFE | Predictors | Correctors |
|---------|------|-----------|------------|------------|
| ct_t100 | 1600 | 128       | 128        | 0          |
| ct_t100 | 1601 | 128       | 128        | 0          |
| ce_t100 | 1600 | 128       | 128        | 0          |
| ce_t100 | 1601 | 128       | 128        | 0          |
| ct_t050 | 1600 | 128       | 128        | 0          |
| ct_t050 | 1601 | 128       | 128        | 0          |
| ce_t050 | 1600 | 128       | 128        | 0          |
| ce_t050 | 1601 | 128       | 128        | 0          |

## Checkpoint logit interpretation

| Config  | Seed | Objective    | Parameterization | Weights |
|---------|------|--------------|------------------|---------|
| ct_t100 | 1600 | CT / raw-LOO | raw_loo          | ema     |
| ct_t100 | 1601 | CT / raw-LOO | raw_loo          | ema     |
| ce_t100 | 1600 | clean CE     | x0_denoiser      | ema     |
| ce_t100 | 1601 | clean CE     | x0_denoiser      | ema     |
| ct_t050 | 1600 | CT / raw-LOO | raw_loo          | ema     |
| ct_t050 | 1601 | CT / raw-LOO | raw_loo          | ema     |
| ce_t050 | 1600 | clean CE     | x0_denoiser      | ema     |
| ce_t050 | 1601 | clean CE     | x0_denoiser      | ema     |

## Per-configuration mean, equal seed weights

*released\_comparable*

| Config  | Seeds complete | validity          | uniqueness        | quality           | diversity         |
|---------|----------------|-------------------|-------------------|-------------------|-------------------|
| ct_t100 | 2/2            | 0.9750 +/- 0.0212 | 0.9948 +/- 0.0074 | 0.4650 +/- 0.0636 | 0.9090 +/- 0.0001 |
| ce_t100 | 2/2            | 0.9800 +/- 0.0141 | 0.9948 +/- 0.0073 | 0.4100 +/- 0.0424 | 0.9025 +/- 0.0036 |
| ct_t050 | 2/2            | 0.9850 +/- 0.0071 | 0.9949 +/- 0.0071 | 0.5450 +/- 0.0212 | 0.8838 +/- 0.0028 |
| ce_t050 | 2/2            | 0.9900 +/- 0.0000 | 0.9949 +/- 0.0071 | 0.5250 +/- 0.0495 | 0.8880 +/- 0.0059 |

*strict*

| Config  | Seeds complete | validity          | uniqueness        | quality           | diversity         |
|---------|----------------|-------------------|-------------------|-------------------|-------------------|
| ct_t100 | 2/2            | 0.3900 +/- 0.0566 | 1.0000 +/- 0.0000 | 0.2100 +/- 0.0141 | 0.8864 +/- 0.0068 |
| ce_t100 | 2/2            | 0.6450 +/- 0.0778 | 1.0000 +/- 0.0000 | 0.2600 +/- 0.0707 | 0.8867 +/- 0.0049 |
| ct_t050 | 2/2            | 0.7700 +/- 0.0141 | 1.0000 +/- 0.0000 | 0.4400 +/- 0.0141 | 0.8683 +/- 0.0082 |
| ce_t050 | 2/2            | 0.6700 +/- 0.0566 | 1.0000 +/- 0.0000 | 0.3950 +/- 0.0354 | 0.8700 +/- 0.0007 |

Every scheduled seed

| Config  | Seed | Status    | Requests | Raw rows | Repair Q | Strict Q | Generation s |
|---------|------|-----------|----------|----------|----------|----------|--------------|
| ct_t100 | 1600 | completed | 100      | 100      | 0.5100   | 0.2200   | 12.24        |
| ct_t100 | 1601 | completed | 100      | 100      | 0.4200   | 0.2000   | 13.86        |
| ce_t100 | 1600 | completed | 100      | 100      | 0.4400   | 0.3100   | 12.40        |
| ce_t100 | 1601 | completed | 100      | 100      | 0.3800   | 0.2100   | 11.93        |
| ct_t050 | 1600 | completed | 100      | 100      | 0.5300   | 0.4500   | 14.36        |
| ct_t050 | 1601 | completed | 100      | 100      | 0.5600   | 0.4300   | 13.98        |
| ce_t050 | 1600 | completed | 100      | 100      | 0.4900   | 0.3700   | 12.52        |
| ce_t050 | 1601 | completed | 100      | 100      | 0.5600   | 0.4200   | 13.92        |

## Preserved v4 diagnostic failure

V4 remains campaign\_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={ 'device': 'cuda:0', 'diffusion\_type': 'udlm', 'exclude\_special\_tokens': False, 'inference\_eps': 1e-05, 'min\_add\_len': 40, 'model\_path': '/home/aidar.alimbayev/Documents/genmolv2/run\_sources/udlm\_genmol\_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num\_samples': 32, 'num\_steps': 128, 'prior\_metadata\_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior\_variant': 'empirical\_frequency', 'randomness': 0.0, 'raw\_loo\_top\_p': 1.0, 'softmax\_temp': 1.0}; expected { 'model\_path': '/home/aidar.alimbayev/Documents/genmolv2/run\_sources/udlm\_genmol\_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num\_samples': 32, 'diffusion\_type': 'udlm', 'softmax\_temp': 1.0, 'randomness': 0.0, 'min\_add\_len': 40, 'num\_steps': 128, 'inference\_eps': 1e-05, 'exclude\_special\_tokens': False, 'prior\_variant': 'empirical\_frequency', 'prior\_metadata\_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective\_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

## Configuration and artifact provenance

Protocol SHA-256: 718893198b0e08a2405e33177c64989499914207ff39b6880edbc5eff2b35c7d

v9-ct-t100 / seed 1600: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 1.0}, "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 1.0}, "gpu": {"cuda\_visible\_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 395138, "process\_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1236}], "index": 7, "memory\_total\_mib": 49140, "memory\_used\_mib": 1259, "name": "NVIDIA RTX A6000", "utilization\_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than"}, "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216}, "physical\_index": "7"}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw\_loo", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ct-t100/seed\_1600/raw\_samples.csv", "sha256": "f8b3994c1a65b0070cf248e44d795a4071ec06f521a7b693b8aba10d1457445d", "size\_bytes": 30618}, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ct-t100 / seed 1601: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 1.0}, "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256":

"f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 1.0}, "gpu": {"cuda\_visible\_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 340770, "process\_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1284}], "index": 2, "memory\_total\_mib": 49140, "memory\_used\_mib": 1307, "name": "NVIDIA RTX A6000", "utilization\_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than"}, "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216}, "physical\_index": 2}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw\_loo", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ct-t100/seed\_1601/raw\_samples.csv", "sha256": "13a78c24b80be0063ee9e44b9aace057d322c35acb2af114efef1f50cb374ee6", "size\_bytes": 29880}, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ce-t100 / seed 1600: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "parameterization": "x0\_denoiser", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 1.0}, "denoiser\_source\_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 1.0}, "gpu": {"cuda\_visible\_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 395138, "process\_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1236}], "index": 7, "memory\_total\_mib": 49140, "memory\_used\_mib": 1259, "name": "NVIDIA RTX A6000", "utilization\_percent": 6, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than"}, "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216}, "physical\_index": 7}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0\_denoiser", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ce-t100/seed\_1600/raw\_samples.csv", "sha256": "7680236fc4efc888b85c91569430d40ad0e44d9337983249957f9fd72a8b27c8", "size\_bytes": 32879}, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm\_denoiser\_metadata": {"inference\_conversion": "subtract\_local\_forward\_log\_likelihood\_before\_controls", "objective": "clean\_token\_cross\_entropy", "parameterization": "x0\_denoiser", "schema\_version": 1}}

v9-ce-t100 / seed 1601: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "parameterization": "x0\_denoiser", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 1.0}, "denoiser\_source\_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true,

"single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 1.0}, "gpu": {"cuda\_visible\_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 343481, "process\_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1256}], "index": 6, "memory\_total\_mib": 49140, "memory\_used\_mib": 1279, "name": "NVIDIA RTX A6000", "utilization\_percent": 0, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than"}, "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216}, "physical\_index": "6"}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0\_denoiser", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ce-t100/seed\_1601/raw\_samples.csv", "sha256": "91de05669cdc5be79830e1c6173a395309ed478941d3e93cbd775556d52c84b3", "size\_bytes": 32925}, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm\_denoiser\_metadata": {"inference\_conversion": "subtract\_local\_forward\_log\_likelihood\_before\_controls", "objective": "clean\_token\_cross\_entropy", "parameterization": "x0\_denoiser", "schema\_version": 1}}

v9-ct-t050 / seed 1600: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 0.5}, "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 0.5}, "gpu": {"cuda\_visible\_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 395138, "process\_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1236}], "index": 7, "memory\_total\_mib": 49140, "memory\_used\_mib": 1259, "name": "NVIDIA RTX A6000", "utilization\_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than"}, "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216}, "physical\_index": "7"}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw\_loo", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ct-t050/seed\_1600/raw\_samples.csv", "sha256": "11dbab23177d6591de03c5097edb9d7f4f7a000554f19906f07c043261cec716", "size\_bytes": 32877}, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}

v9-ct-t050 / seed 1601: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 0.5}, "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema"}, "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 0.5}, "gpu": {"cuda\_visible\_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 343481, "process\_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1256}], "index": 6, "memory\_total\_mib": 49140, "memory\_used\_mib": 1279, "name": "NVIDIA RTX A6000", "utilization\_percent": 6, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus",

"max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than", "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216, "physical\_index": 6}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema", "objective": "CT / raw-LOO", "parameterization": "raw\_loo", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ct-t050/seed\_1601/raw\_samples.csv", "sha256": "13d0d3fb32634f20490389e74883f8ba88a196bb8cf565bdca0b5c8f0d9c028e", "size\_bytes": 33609, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9"}}

v9-ce-t050 / seed 1600: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "parameterization": "x0\_denoiser", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 0.5}, "denoiser\_source\_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema", "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 0.5}, "gpu": {"cuda\_visible\_devices": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 395138, "process\_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1236}], "index": 7, "memory\_total\_mib": 49140, "memory\_used\_mib": 1259, "name": "NVIDIA RTX A6000", "utilization\_percent": 0, "uuid": "GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than", "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216, "physical\_index": 7}, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema", "objective": "clean CE", "parameterization": "x0\_denoiser", "raw": {"relative\_path": "output/udlm/engineering\_v9/v9-ce-t050/seed\_1600/raw\_samples.csv", "sha256": "32b342171299c7bfd745c16007ade327ffdb67b94c14047749f140898356c205", "size\_bytes": 32079, "sampling\_budget": {"corrector\_steps\_per\_molecule": 0, "gibbs\_corrector": false, "nfe": 128, "predictor\_transitions\_per\_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm\_denoiser\_metadata": {"inference\_conversion": "subtract\_local\_forward\_log\_likelihood\_before\_controls", "objective": "clean\_token\_cross\_entropy", "parameterization": "x0\_denoiser", "schema\_version": 1}}}}

v9-ce-t050 / seed 1601: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "min\_add\_len": 40, "num\_steps": 128, "parameterization": "x0\_denoiser", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "raw\_loo\_top\_p": 1.0, "softmax\_temp": 0.5}, "denoiser\_source\_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation\_protocol": {"diffusion\_type": "udlm", "exclude\_special\_tokens": false, "inference\_eps": 1e-05, "inference\_weights": {"ema": {"decay": 0.9999, "num\_updates": 1000, "shadow\_parameter\_count": 230}, "ema\_applied": true, "source": "ema", "model\_use\_bracket\_safe": false, "nfe": 128, "nfe\_definition": "one full backbone forward evaluation per reverse step", "num\_steps": 128, "num\_steps\_source": "explicit UDLM reverse-transition count", "prior\_metadata\_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior\_variant": "empirical\_frequency", "randomness": 0.0, "randomness\_used\_by\_sampler": false, "raw\_loo\_top\_p": 1.0, "released\_largest\_component": "maximum SMILES string length", "released\_safe\_fix": true, "single\_generation\_batch": true, "strict\_safe\_fix": false, "temperature": 0.5}, "gpu": {"cuda\_visible\_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final\_probe": {"compute\_mode": "Default", "compute\_processes": [{"pid": 343481, "process\_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO\_repro/bin/python", "used\_memory\_mib": 1256}], "index": 6, "memory\_total\_mib": 49140, "memory\_used\_mib": 1279, "name": "NVIDIA RTX A6000", "utilization\_percent": 6, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch\_policy": {"active\_compute\_processes\_allowed": true, "inventory\_scope": "all\_nvidia\_gpus", "max\_utilization\_percent": 10, "min\_free\_memory\_mib": 30000, "requested\_gpu\_count": 2, "selection\_method": "dynamic\_idle\_discovery", "utilization\_comparison": "strictly\_less\_than", "logical\_device": {"compute\_capability": [8, 6], "logical\_index": 0, "name": "NVIDIA RTX A6000", "total\_memory\_bytes": 50897289216, "physical\_index": 7}}}}



```
"6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE",  
"parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/engineering_v9/v9-ce-t050/seed_1601/raw_samples.csv", "sha256":  
"41af77517633bd59388930f20f0435a3b57d86cb354a2e1b9ec5095a7f4e98dd", "size_bytes": 33179}, "sampling_budget": {"corrector_steps_per_molecule": 0,  
"gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "da8f5403ec9de02792261d1ffc3aa20ac08474e9", "udlm_denoiser_metadata":  
{"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser",  
"schema_version": 1}}
```